LIGHTS OUT

Project Overview

Duration: 4–6 weeks
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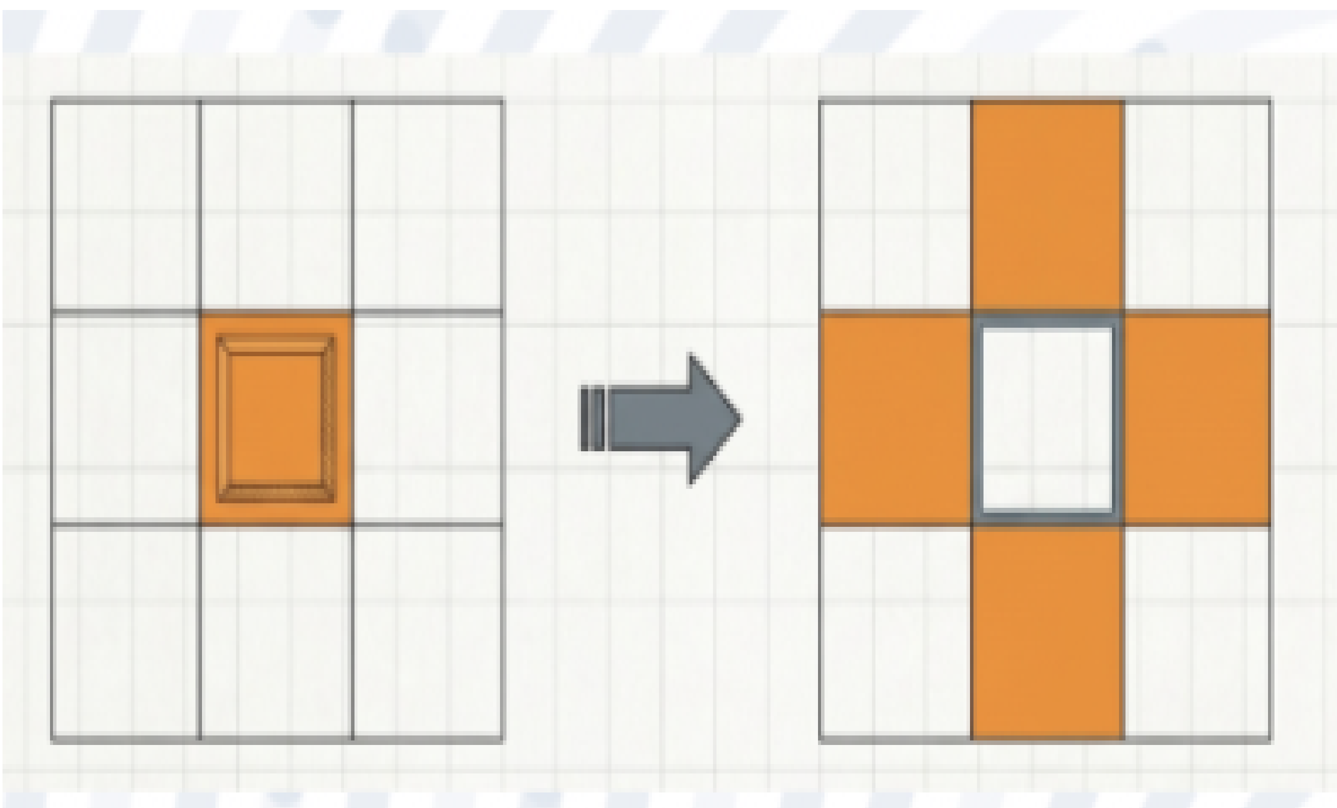
This project presents the design and implementation of a Lights Out puzzle game on an FPGA platform using Verilog HDL. The system uses VGA output to display an interactive 8×8 grid and push-button inputs for user control. The objective of the game is to switch off all lights by toggling selected cells, where each toggle affects adjacent cells.



Introduction

1.1 Project Statement

The objective of this project is to develop a hardware-based interactive gaming system using FPGA. The system must generate VGA signals, display a gridbased interface, process user inputs through buttons, and implement game logic to detect completion.

1.2 Background Survey

Field Programmable Gate Arrays (FPGAs) are widely used for real-time digital systems due to their parallel processing capability and configurability. The Lights Out game is a well-known puzzle where pressing a cell toggles its state along with its neighboring cells. Implementing such a system on FPGA provides practical exposure to digital design concepts such as FSM, synchronous design, and hardware interfacing.

Methodology

2.1 Techniques & Approach

The system is designed using a modular approach. A 25 MHz clock is derived from the main clock to support VGA timing. Push-button inputs are debounced to eliminate noise. The game logic is implemented using a finite state machine and a 64-bit register representing the grid. VGA signals are generated to display the grid, and pixel rendering logic determines the visual output.

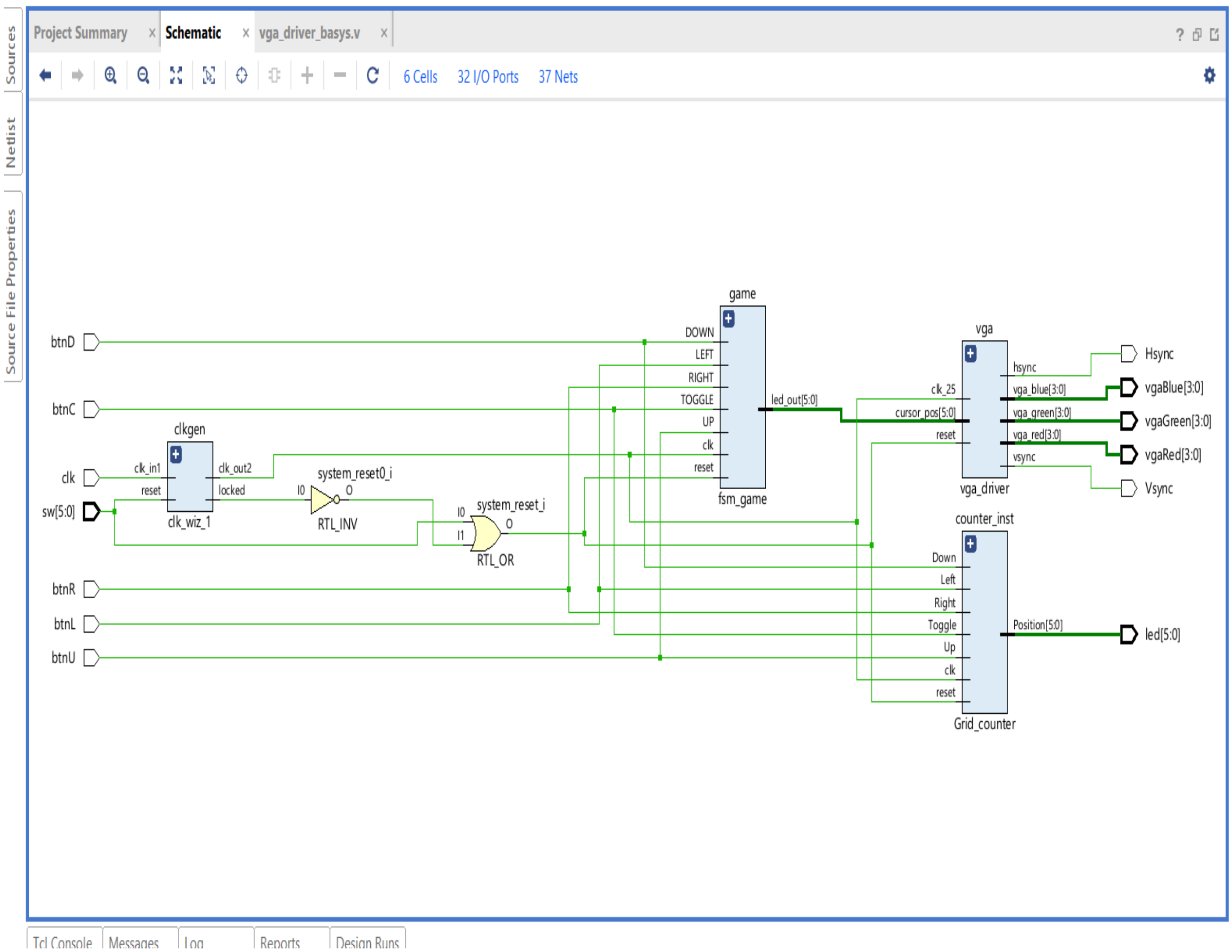
2.2 System Architecture & Workflow

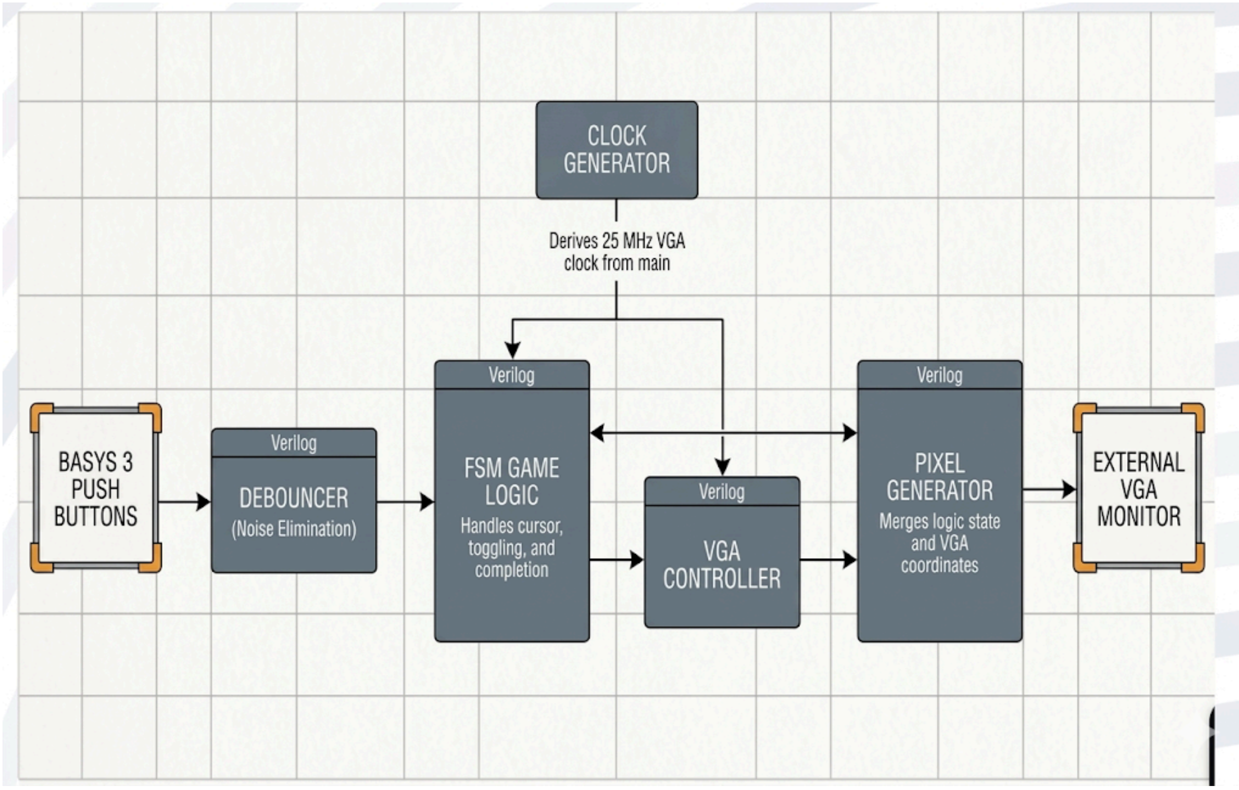
The system consists of multiple interconnected modules. The top module integrates all components. A clock generator produces the required VGA clock. Debounce modules ensure clean button inputs. The game logic module controls cursor movement, toggling operations, and completion detection using a finite state machine. The VGA controller generates synchronization signals and pixel coordinates. The pixel generator maps the game state to visual output, displaying the grid, active cursor, and light status.

2.3 Tools & Technologies Used

The project is implemented using Verilog HDL and developed using Xilinx Vivado. The design is deployed on an FPGA board Basys 3 and displayed using a VGA monitor.

Block Diagrams





Results & Analysis

The implemented system successfully displays an 8×8 grid on a VGA screen and allows real-time interaction through push buttons. The toggling logic correctly updates the selected cell and its adjacent neighbors. The cursor is visibly highlighted, and the system accurately detects when all lights are turned off, indicating completion. The VGA output is stable and operates at the standard 640×480 resolution.

Challenges & Learnings

4.1 Obstacles Faced

Several challenges were encountered during the implementation. Button bouncing caused multiple unintended inputs. Synchronizing VGA signals required precise timing. Representing a two-dimensional grid using a one-dimensional array added complexity. Debugging state transitions in the finite state machine was also challenging.

4.2 Solutions & Insights

Debouncing logic with a delay mechanism was implemented to eliminate noise from button inputs. Modular design helped isolate and debug issues effectively. The grid was mapped using index calculations, simplifying access to individual cells. The project provided valuable insights into timing constraints, hardware debugging, and real-time system design.

Conclusion & Applications

5.1 Final Takeaways

This project demonstrates the successful implementation of an interactive game using an FPGA. It highlights the integration of digital design concepts such as FSM, VGA interfacing, and synchronous logic. The system performs reliably and efficiently, showcasing the capabilities of hardware-based design.

5.2 Real-World Applications

The concepts used in this project can be applied in embedded gaming systems, digital display controllers, educational tools, and interactive hardware Interfaces.

Future Work & Enhancements

Future improvements may include adding multiple difficulty levels, implementing a scoring system, introducing a timer, enhancing graphics, and supporting alternative input methods such as keyboards. The grid size can also be expanded for increased complexity.

References

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